Unlocking the Mathematical Code Behind Cybersecurity

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Abstract

Cybersecurity is no longer limited to Software Developers or government agencies; it concerns every citizen who uses a smartphone, accesses social media, or stores information online. As India shifts into the digital era, the risks to personal and institutional data have skyrocketed. This paper aims to **demystify the mathematical foundations of cybersecurity**, making them understandable for high schoolers and educators, and **highlights the urgent need for basic cybersecurity awareness** in Indian classrooms. Through models, mathematical tools like **number theory, probability, modular arithmetic, and cryptographic algorithms** have been simplified. Without going into much detail of the math, this paper aims to present a clear-cut awareness-based discussion of the security segment.

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1. Introduction

With more than **850 million internet users in 2025**, India has transformed into one of the largest digital populations in the world. From healthcare to education, every sector has empowered technology. But while access has grown, digital safety has not kept pace. According to **CERT-In**, **over 1.4 million cybercrimes** were reported in 2023 alone. Students, staff members, and educators are increasingly becoming soft targets.

Cybersecurity is often seen as a job for engineers, coders, or more generally, nerds. However, at its core, cybersecurity depends on mathematics. Encryption, password protection, and secure online communication rely on sheer mathematical logic. Simultaneously, **cyber awareness is extremely low** among students and teachers. This paper focuses on both sides of the coin -:

- ➢ How does mathematics change the entire understanding of cybersecurity?
- ➢ Why is awareness in schools an alarming call to protect India's digital future?

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2. Mathematics: The Dark Knight

2.1 Number Theory and Encryption

One of the most powerful tools in cybersecurity is **encryption**. It scrambles data into unreadable code, so that only the intended person can read it. The popular **RSA algorithm** (used in banking, websites, and emails) relies on a simple mathematical truth:

> It's easy to multiply two large prime numbers, but very hard to figure them out if we only know the result. (Based on Fermat's theorems)

RSA encryption is based on this idea. Here's how it works:

Step 1: We choose two large prime numbers (p and q),
Step 2: Multiply them to get $\mathbf{n = p \times q}$,
Step 3: Then use the formula:
$\mathbf{C \equiv M^e \bmod n}$
where **C = encrypted message**, **M = original message**, and **e** is a number used for encryption.

This sounds complicated, but it means that only someone with the right key can unscramble the message.

To understand this better, imagine writing a message in a secret language only your best friend knows. Even if someone else finds the message, they won't be able to understand it without knowing the language rules. Encryption works the same way; it locks information using math so that only the person with the correct key can unlock it.

RSA is trusted globally because of how tough it is to break. Even supercomputers would take years or maybe centuries to crack an RSA-encrypted message without the key. This is why it's used in ATMs, online shopping, email security, and even official communications.

Most people use RSA every day without realizing it. When we see a website that begins with "https", it means our browser and the website are using encryption (like RSA) to protect what we type, whether it's our password or our credit card number.

So while the math behind RSA may seem hidden in the background, it is quietly working to keep our online world safe.

2.2 Hashing and Modular Arithmetic

A hash function takes any amount of data and converts it into a short, fixed string of numbers and letters. For example, passwords are often stored as hashes. If two different passwords gave the same hash, the system could be tricked; this is called a collision.
To prevent this, modern hash functions like **SHA-256** use modular arithmetic—which involves remainders after division—to mix the data so well that no two inputs create the same hash.
Hashing is used in:

1. Storing passwords safely,
2. Ensuring file integrity,

3. Blockchain technology is like cryptocurrencies.

Now let's break this down.

Imagine we write our password on a piece of paper. That's not safe; someone could read it. So instead, our computer uses a trick: it turns our password into a secret code, a kind of fingerprint, that can't be reversed. That's what a **hash** is. Even if someone sees the hash, they can't tell what our password is. It's like looking at a cake and trying to figure out the exact ingredients used.

Also, if someone tries to log in using a fake password, the computer will hash that password too, but the result will look different. That's how the system knows it's wrong. If two different passwords gave the same hash, the system would get confused. That's why **"collisions"** (two different inputs giving the same output) are dangerous, and this is why hash functions are carefully designed to avoid them completely.

Think of hashing like blending fruits into a smoothie. Once blended, it's almost impossible to tell exactly what went in. But if we always blend the same fruits in the same amounts, we'll always get the same smoothie. Changing even one fruit will result in a change of both taste and colour of the smoothie. That's how hashing works.

Hashing is everywhere around us:

- When we create a password for a social media account, the company doesn't store our actual password. They store its hash, so even if their data is hacked, our real password is still safe.
- When files are sent over the internet, hashes help check if the file has been tampered with or changed along the way.
- In cryptocurrencies like Bitcoin, hashing helps keep track of who owns what and ensures that no one cheats the system.

In short, hashing is like sealing our digital data with a unique stamp that stays sealed and proves that it hasn't been changed.

2.3 Miscellaneous Models

2.3.1 Probability and Threat Analysis

One example of cybercrime is **brute-force attacks**, where a hacker tries every possible password until they get the right one.

If our password has 8 characters (using uppercase, lowercase, numbers, and symbols = 94 characters), then:

Total combinations = $94^8 \approx 6.1 \times 10^{15}$ (using basic principles of combination of numbers when repetition is allowed)

Even with supercomputers, trying all combinations consumes time. So, math helps cybersecurity experts to get an idea about risks and build stronger defense systems.

2.3.2 Cryptographic Algorithms and Linear Algebra

Advanced algorithms like **AES (Advanced Encryption Standard)** are used by companies and top-secret agencies to protect important data. AES uses:

- Matrix multiplication from linear algebra.
- Permutation and Combination to scramble data.

For example, **AES-256** (used for military and financial security) runs data through **14 rounds** of these operations. All of this is based on mathematical rules, not just programming.

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3. Awareness: The Missing Link

3.1 Human Error in Cyber Attacks

A shocking fact: **95% of cyberattacks occur due to human error** (IBM, 2023).
Examples include:

- Students using weak passwords like “123456”, “abcdefg”, “<date of birth/fictional character>”, etc.

- Teachers clicking on phishing emails asking for personal/financial info.

Even the best technology cannot protect those users who are not simply aware.

3.2 Case Study: 2022 EdTech Data Breach

In 2022, a major Indian EdTech company lost the personal data of **over 2 million students and teachers**. This included names, email addresses, phone numbers, and login details. It was one of the **biggest education-related data breaches in India’s history**.

Why did this happen?
It wasn’t because of a deadly hacker or system failure. Unfortunately, the reason was much simpler:

- **Weak passwords:**
  Many users were using easy passwords like “password123” or their names. These are the first things a hacker will try. It’s like locking our house with a plastic toy lock that even a

10-year-old can break.

- **No two-factor authentication (2FA):**
  Two-factor authentication adds a second layer of security. For example, after entering our password, we might get an OTP on our phone to confirm it's us. But this company didn't make it compulsory. That meant anyone who guessed or stole a password could easily access the account, with no extra security measures in place.

- **No awareness among users:**
  Teachers and students using the platform didn't know the basic rules of staying safe online. Many didn't recognize fake emails or messages asking for their passwords. Some even shared their login details by mistake. This lack of knowledge made it easier for attackers to trick users and get into the system.

This case shows how cybersecurity failures are not always about technology; they're often about people. If the company had trained its users, enforced strong password policies, and added basic protections like 2FA, the breach could have been prevented.

What's worse is that the affected students and teachers may not have even known their data was exposed. That's why **cyber awareness** is just as important as software development and design. Every student, teacher, and school must understand these risks because once data is leaked, we can't get it back.

This breach could have been easily prevented with basic training.

3.3 Awareness Gaps in Indian Classrooms

A **2024 DSCI survey** found:

- Only **18%** of students had basic cybersecurity training,

- Only **24%** of teachers knew how to protect themselves online.

Most schools in India do **not teach students about safe internet habits**, recognizing phishing emails, or how to protect digital identities.

3.4 Best Practices for Students and Educators

Here are some efficient techniques that can prevent most cyberattacks:

- Use **strong passwords** (mix letters, numbers, special characters).

- Try to maintain **unique passwords** for different platforms..

- Enable **2-factor authentication** (like OTP on phone).
- Avoid clicking on unknown or suspicious links.
- Keep software and apps updated.
- Use **trusted platforms** for personal and professional use.

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4. Statistical Evidence and Suggestions

4.1 Cybercrime Trends in the Educational Sector

Year	Total Cyber Incidents	In the Education Sector
2021	1.2 million	3,000
2022	1.3 million	7,500
2023	1.4 million	12,000

(Source: CERT-In Annual Reports)

This sharp increase shows that schools are becoming regular targets of cybercrime. If ignored, it could have a fatal outcome on exams and student data security.

4.2 What the Government Is Doing

The Indian government has launched programs to fight cyber threats:

- **Cyber Swachhta Kendra**: Gives tools to remove viruses and protect devices.
- **Information Security Education and Awareness (ISEA)**: Offers training and workshops on safe online behavior.

However, these programs are not yet reaching most schools. There's an **urgent need** to:

- Integrate cybersecurity lessons into school curricula,
- Teach these concepts during math and computer classes,
- Conduct workshops for teachers and students together.

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5. Conclusion

Cybersecurity is more than a buzzword; it's a basic necessity for survival. This paper has shown how deeply mathematics is involved in protecting our digital lives, through encryption, hashing, and algorithms. But this simply is not enough.

Unless students and teachers understand the risks and adopt safe digital habits, even the best security can fail. India's journey to becoming a digital messiah depends not just on infrastructure, but on cyber-aware citizens.

Last, but not least,

> *Math protects us. Awareness empowers us. We need both.*

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